

Lab assignment 4 - Local features matching and Harris corner detection Complete

3.11.2025

Attempt 1



Review Feedback

20.10.2025

Attempt 1 Score:

Complete



Add Comment

Anonymous Grading: **no**

Unlimited Attempts Allowed

22.9.2025 to 17.11.2025

Details

Lab assignment 4 (individual)

In this individual lab assignment, every student must complete the following tasks.

The code created for the exercises has to be delivered to the previously created repository.

The output images are delivered together in a PDF which needs to be delivered here on Canvas.

I. Assignment directory:

1. Create a new project directory "assignment_4" inside your "IKT213_lastname" repository

II. Create functions to find key features matching between two images (using SIFT or ORB or SURF) and if matching was found, then align the images.

1. Harris Corner Detection: Implement a function which receives an image, detects its edges, and saves the image with the marked edges

to learn more, go to "https://opencv24-python-tutorials.readthedocs.io/en/latest/py_tutorials/py_feature2d/py_features_harris/py_features_harris.html"  https://opencv24-python-tutorials.readthedocs.io/en/latest/py_tutorials/py_feature2d/py_features_harris/py_features_harris.html."

A. Function argument: **reference_image**

B. Use "[reference_img.png](https://uia.instructure.com/courses/18389/files/3039785?wrap=1) (<https://uia.instructure.com/courses/18389/files/3039785?wrap=1>)  (https://uia.instructure.com/courses/18389/files/3039785/download?download_frd=1) " for the "reference_image".

C. Save the image: harris (this is page 1 of PDF)

2. Feature-Based Image Alignment: Your task is to be able to align objects in an image based on the detected edges of a given reference image

(Use **ONE** of the approaches below, leave a note in the submission on which one you used)

2.1. Using SIFT:

"[OpenCV: Feature Matching + Homography to find Objects](https://docs.opencv.org/4.x/d1/de0/tutorial_py_feature_homography.html)  (https://docs.opencv.org/4.x/d1/de0/tutorial_py_feature_homography.html)."

2.2. Using SURF:

"[OpenCV: Feature Matching with FLANN](https://docs.opencv.org/4.x/d5/d6f/tutorial_feature_flann_matcher.html)  (https://docs.opencv.org/4.x/d5/d6f/tutorial_feature_flann_matcher.html)."

2.3. Using ORB (warning: not very stable, but feel free to try):

"[Feature Based Image Alignment using OpenCV \(C++/Python\) \(learnopencv.com\)](https://learnopencv.com/image-alignment-feature-based-using-opencv-c-python/)  (<https://learnopencv.com/image-alignment-feature-based-using-opencv-c-python/>). " &
"<https://github.com/spmallick/learnopencv/blob/master/ImageAlignment-FeatureBased/align.py>  (<https://github.com/spmallick/learnopencv/blob/master/ImageAlignment-FeatureBased/align.py>). " "

- A. Function arguments: `image_to_align`, `reference_image`, `max_features`, `good_match_precent`
- B. Use "[align_this.jpg \(https://uia.instructure.com/courses/18389/files/3039787?wrap=1\)](https://uia.instructure.com/courses/18389/files/3039787?wrap=1)  (https://uia.instructure.com/courses/18389/files/3039787/download?download_frd=1) " for the `"image_to_align"` and "[reference_img.png \(https://uia.instructure.com/courses/18389/files/3039790?wrap=1\)](https://uia.instructure.com/courses/18389/files/3039790?wrap=1)  (https://uia.instructure.com/courses/18389/files/3039790/download?download_frd=1) " for the `"reference_image"`.
 - Use `max_feature=1500`, and `good_match_precent=0.15` (for ORB)
 - Use `max_features=10`, and `good_match_precent=0.7` (for SIFT/FLANN solution)
- C. Save the two images: `aligned` and `matches` (this is page 2 and page 3 of PDF)

PS! THE SORT FUNCTION FROM ORB EXAMPLE HAS AN ERROR! FIX:

replace this `"matches.sort(key=lambda x: x.distance, reverse=False)"`

with this `"matches = sorted(matches, key=lambda x: x.distance, reverse=False)"`

Useful links:

- [A Detailed Guide to the Powerful SIFT Technique for Image Matching \(with Python code\)](https://www.analyticsvidhya.com/blog/2019/10/detailed-guide-powerful-sift-technique-image-matching-python/)  (<https://www.analyticsvidhya.com/blog/2019/10/detailed-guide-powerful-sift-technique-image-matching-python/>)
- [Image stitching tutorial OpenCV](https://github.com/linrl3/Image-Stitching-OpenCV)  (<https://github.com/linrl3/Image-Stitching-OpenCV>)
- [Image stitching code example](https://github.com/pavanpn/Image-Stitching)  (<https://github.com/pavanpn/Image-Stitching>)

Do you need help in the lab or in Discord? Find links here: [Lab Information \(https://uia.instructure.com/courses/18389/pages/lab-information\)](https://uia.instructure.com/courses/18389/pages/lab-information)

